

Digital Noise Generator (DNG)

Selecting suitable noise transistors—a task that cannot be avoided when building conventional noise generators—can easily turn into a time-consuming and costly undertaking. It is far simpler (and no more expensive) to use a digital noise generator. Therefore, such a circuit should no longer be withheld from the FORMANT player.

For the acoustician, white noise is what white light is to the optician. It is composed of the harmonic oscillations of all frequencies within the audible range and can be generated electronically in a variety of ways.

In this case, the noise signal is derived not from transistor junction breakdown, but from the output of a digital 18-bit shift register (Figure 1). The shift register receives its control signals from a high-frequency clock oscillator (clock frequency approx. 30 kHz) built around N1 and N2. A feedback loop incorporating N3, T1, and N4 is used to mix the circulating digital signals. This creates a continuous sequence of "0"s and "1"s that is fed back into the circuit. Unlike an analog generator, which produces a random noise pattern, the signals here repeat with a period of approximately 1 second; however, this repetition is inaudible. IC3 adjusts the signal to the standard level used in the FORMANT system (approx. 2.5 V). It goes without saying that the supply voltage for the CMOS ICs is limited to approximately -6.3 V to -7.3 V by D1, allowing even unbuffered types to be used safely.

Construction and Assembly

The suggested PCB layout and component placement diagram are shown in Figure 2.

As this is the first FORMANT circuit to use CMOS ICs, it is worth noting that the standard precautions for handling CMOS components must be strictly observed. However, if you prefer to solder the ICs directly onto the board, you should solder the power supply pins (Pin 7, Pin 14) first so that the internal protection diodes can become effective. Otherwise, there are no special considerations regarding the assembly of the digital noise generator.

As illustrated in Figure 3, the circuit board can be mounted to the FORMANT noise board using 20–30 mm spacers. The output of the DNG must be connected to the WN output on the noise board and the front panel. Due to its linear frequency response, the DNG is also readily suitable for audio-frequency measurement purposes. The clock frequency is 30 kHz, the period of the digital signal is approx. 1 second, and the RMS voltage of the output signal is approx. 0.9 V.

Figure 4 shows the WN output signal waveform, while Figure 5 shows the pin assignments for the CMOS ICs used.

Parts List for Figure 1

Resistors (carbon film, 5%):

R1 = 47k
R2 = 4.7k
R3 = 150k
R4 = 22k
R5 = 12k
R6...R8 = 39k

Capacitors:

C1 = 100pF
C2 = 33pF
C3 = 1pF
C4 = 1μF/25V (tantalum)

Semiconductors:

T1 = TUN
D1 = BZX 83 C 8V2
N1...N4 = IC1 = CD 4030
IC2 = CD 4006
IC3 = uA 741C (Mini-DIP)

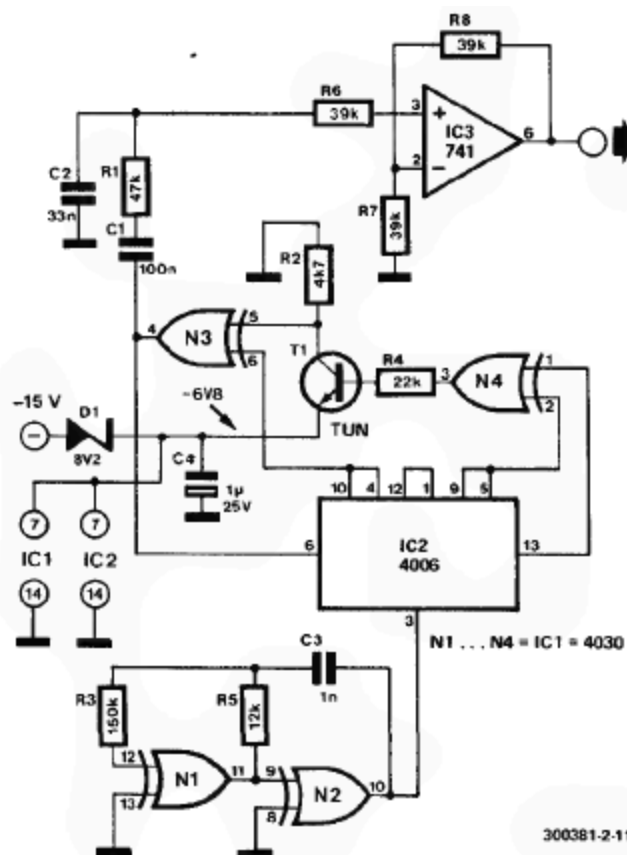
Miscellaneous:

2 spacer sleeves

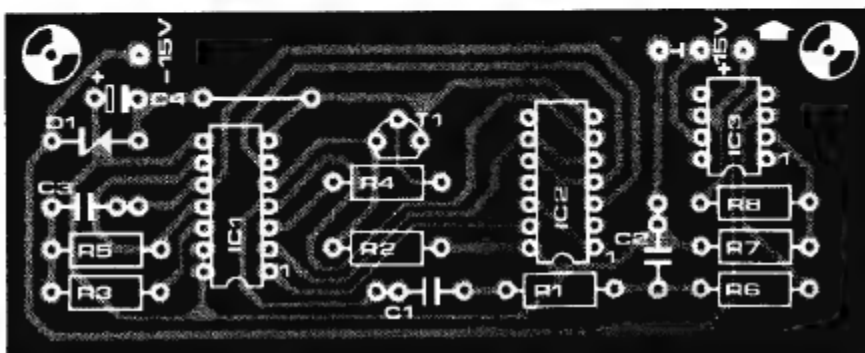
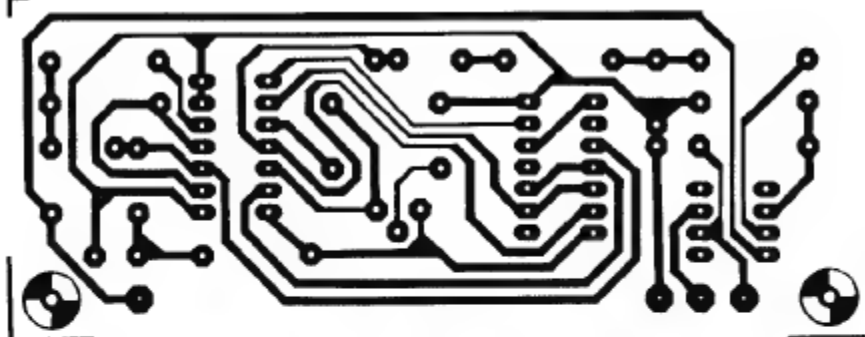
Figures

- Figure 1. Schematic diagram of the digital noise generator.
- Figure 2. Layout and component placement plan for the circuit shown in Figure 1.
- Figure 3. Assembly suggestion and wiring instructions for the auxiliary circuit.
- Figure 4. Signal voltage at the WN output.
- Figure 5. Pinout of the CMOS ICs used.

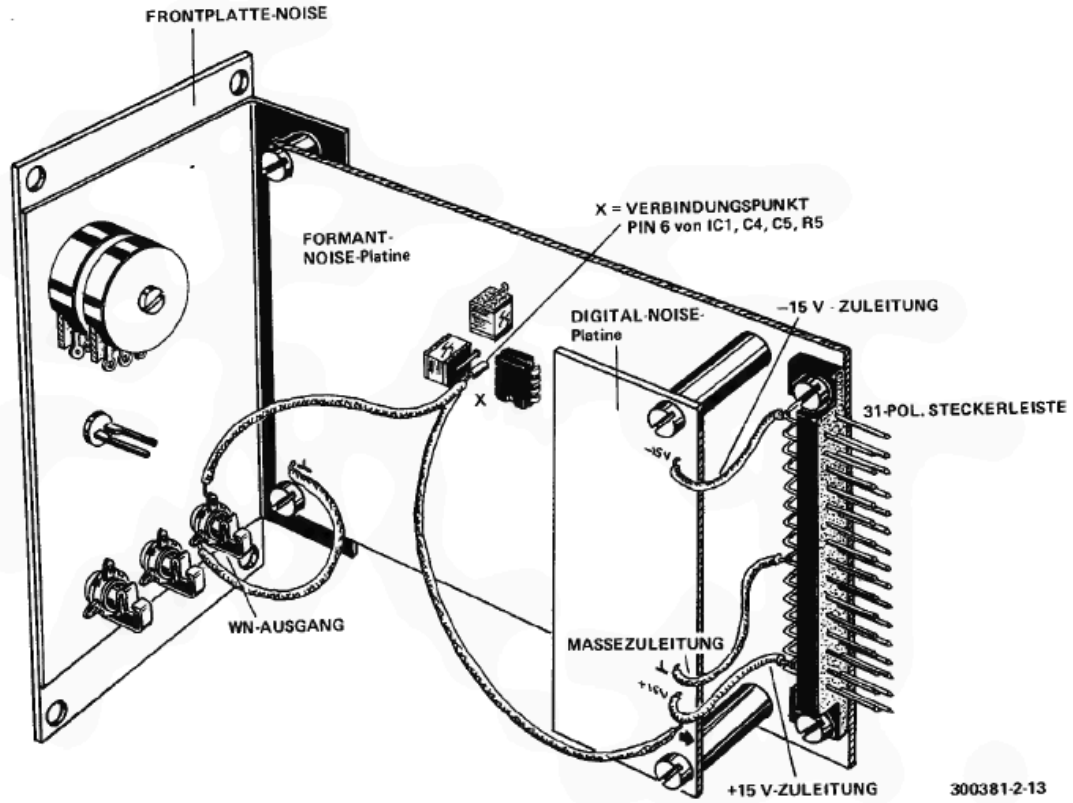
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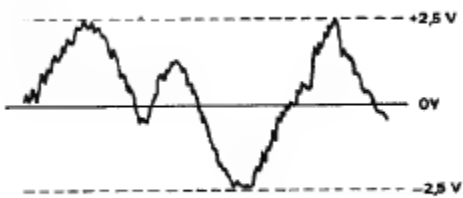


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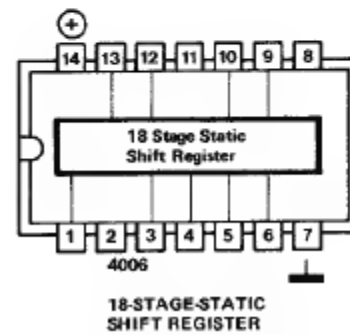
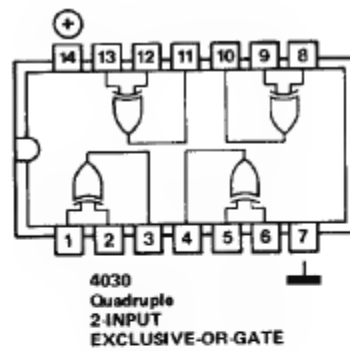
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